ARCHITECTURE WHITEPAPER · VERSION 2.0.0

Institutional Algorithmic Trading Engine

A zero-cost, 24/7 serverless edge architecture for multi-asset momentum breakouts and order block invalidation

ASSET UNIVERSE
EURUSD · USDJPY · USDCAD · USDCHF · AUDUSD · XAUUSD

INFRASTRUCTURE
GitHub Actions Cron + Cloudflare Global Edge

CLIENT INTERFACE
Telegram Bot API (mobile & desktop)

RECURRING COST
$0.00 / month · 100% serverless

RISK POLICY
Zero martingale · zero grid · zero averaging down

FORWARD-AUDIT EPOCH
September 28, 2026

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DOCUMENT SCOPE

This document specifies the design, risk framework, operating model and verification standard of the engine at version 2.0.0. It contains no historical performance figures. Forward results begin accruing on September 28, 2026.

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SECTION 01

Vision

Systematic trading discipline should not depend on costly infrastructure, opaque models, or an unverifiable record.

The Institutional Algorithmic Trading Engine rests on one proposition: the rules, the risk, and the record of a trading system can all be made explicit, inspectable and continuously operational, at zero recurring infrastructure cost.

Conventional retail and prop-firm automation is held back by three structural weaknesses: dependence on always-on local hardware, recurring platform and hosting fees, and strategies tuned to historical data rather than tested against live order flow. This engine is designed around removing each one.

Always on, hardware free

Continuous surveillance of six instruments from serverless cloud components. No local machine, VPS or paid charting tier is required.

Risk before return

Every position carries a structural stop, a fixed 1 : 1.75 target and 1.0% equity risk. Recovery techniques are excluded by design.

Verifiable by construction

Results are reported only from live forward executions, held in a version-controlled ledger with a commit timestamp for every record.

Design objectives

Objective	How the design addresses it
Operational continuity	Scheduled cloud compute replaces desktops and VPS hosting; state lives in the repository, not on a machine.
Explainable decisions	Every signal is the product of four published entry conditions and a deterministic risk construction. There is no hidden model.
Bounded downside	Fixed fractional risk, an 18.0 to 60.0 pip stop range and a one-trade-per-pair lock cap exposure at every level.
Honest measurement	No backtests are published. Metrics are computed strictly from trades taken on or after 2026-09-28.

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SECTION 02

Scope & Boundaries

The engine is a signal generation, monitoring and journaling system. The boundaries below define what it does and what it deliberately does not.

IN SCOPE	OUT OF SCOPE
▪ Six instruments: EURUSD, USDJPY, USDCAD, USDCHF, AUDUSD, XAUUSD ▪ 15M and 1H timeframes, evaluated on confirmed bar closes ▪ Rule-based entry signals with structural stop and fixed target ▪ Exact position sizing for 1.0% equity risk ▪ Trade lifecycle monitoring with a single-trade lock per pair ▪ Atomic trade ledger with git-backed history ▪ Telegram operator interface: radar, prices, history, performance ▪ Forward-only performance reporting	▪ Martingale, grid or averaging-down techniques ▪ Instruments outside the six-asset universe ▪ Published backtests or hypothetical performance ▪ Custody of client funds ▪ Discretionary override of engine rules ▪ Paid infrastructure dependencies ▪ Broker order placement: signals are delivered to the operator

Operating envelope

Parameter	Specification
Asset universe	EURUSD · USDJPY · USDCAD · USDCHF · AUDUSD · XAUUSD (spot gold)
Timeframes	15-minute and 1-hour candles
Scan cadence	Every 15 minutes, Monday to Friday (`*/15 * * * 1-5`)
Market data	Yahoo Finance real-time API (OHLCV)
Risk per trade	1.0% of equity
Reward-to-risk	Fixed 1 : 1.75
Concurrency	One open trade per pair; up to six across the universe
Recurring cost	$0.00 per month

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SECTION 03

Problem & Solution Overview

Three recurring constraints limit retail and prop-firm algorithmic trading. The engine addresses each with a specific design decision.

Constraint	Impact	Design response
Fragile local dependencies	Desktops must stay powered 24 hours a day, exposed to power cuts, Wi-Fi drops and OS updates.	Scheduled cloud compute with no local machine in the loop.
Recurring subscriptions	VPS hosting ($200 to $500 per year) and TradingView Premium ($600+ per year) for continuous alerts.	Free-tier serverless components across the full pipeline.
Over-optimized black boxes	Curve-fitted strategies that excel in backtests but degrade against live order flow.	Simple, published rules and forward-only measurement.

The pipeline is split across three serverless components: **GitHub Actions** for scheduled compute, **Cloudflare Global Edge Workers** for webhook handling and state, and the **Telegram Bot API** as the operator interface. The result is continuous market surveillance, structural risk management and a forward-verified trade journal.

$0.00
Monthly infrastructure cost

24 / 5
Surveillance, every 15 minutes

1 : 1.75
Fixed risk-to-reward

1.0%
Equity risked per trade

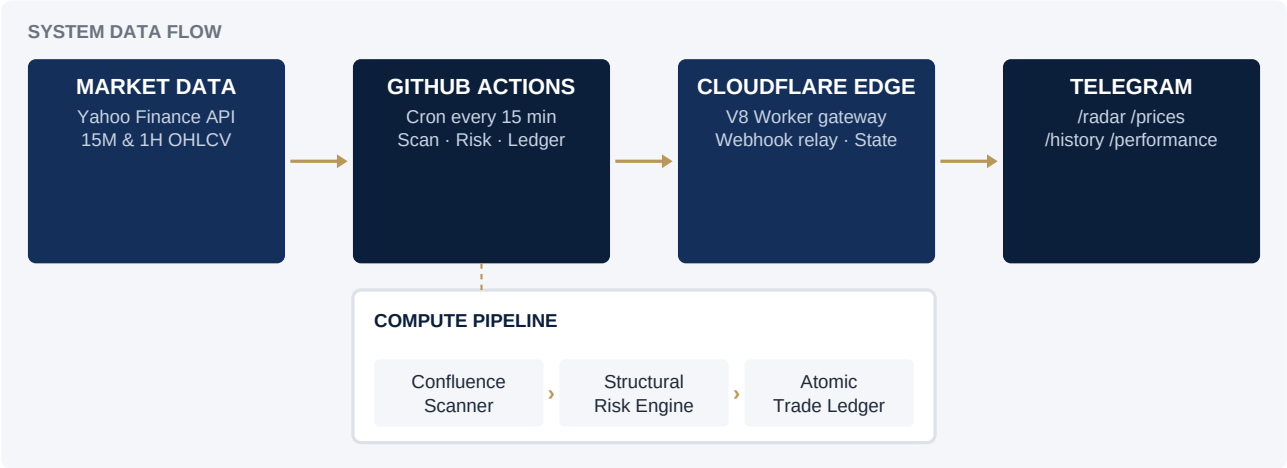
CORE RISK INVARIANT
The system operates with strictly **zero martingale, zero grid recovery and zero averaging down**. Every trade is bound to an immutable structural Stop Loss and a fixed 1 : 1.75 Take Profit.

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SECTION 04

System Architecture

The architecture is built from asynchronous, event-driven components running independently across global datacenters. There is no single point of failure and no dependence on a local machine.



Layer	Technology	Responsibility
Market Feed	Yahoo Finance Real-Time API	Supplies 15M and 1H OHLCV candles for Forex majors and spot gold.
Surveillance Engine	GitHub Actions Cloud Cron	Runs scan cycles every 15 minutes; evaluates breakouts and active-trade TP/SL hits.
Trade Ledger	`trade_records.json` + Git	Maintains atomic state of open and completed trades; auto-commits exit resolutions.
Edge Gateway	Cloudflare Workers (V8)	Sub-50 ms gateway; manages Telegram webhooks, command routing and memory state.
Client Interface	Telegram Bot (mobile / desktop)	Tap-to-copy signals, market radar telemetry and filterable trade history.

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SECTION 05

Quantitative Strategy Mechanics

The engine merges classical **Donchian Channel momentum breakouts** with **Smart Money Concept (SMC) Order Block** invalidation. Signals are evaluated on completed, confirmed bar closes only. Conditions 1 to 4 must hold together to qualify an entry; conditions 5 and 6 define the trade's risk structure.

1	**Structural 20-period breakout** Close must be strictly outside the 20-period highest-high / lowest-low channel.	`Long: Close > max(High[t-1 … t-20])` `Short: Close < min(Low[t-1 … t-20])`
2	**Multi-EMA trend & velocity slope** Filters sideways chop and counter-trend traps. Long shown; invert for shorts.	`Close > EMA50 · EMA20 > EMA50` `EMA20[t] − EMA20[t-1] > 0`
3	**Anti-exhaustion RSI window** Avoids entries at the climax of parabolic thrusts.	`Long: 45.0 ≤ RSI(14) ≤ 75.0` `Short: 25.0 ≤ RSI(14) ≤ 55.0`
4	**Origin Order Block anchor** Institutional footprint immediately preceding displacement: last down-candle for longs, last up-candle for shorts.	`Anchor = last opposing candle before breakout`
5	**Structural invalidation (Stop Loss)** Placed beyond the Order Block extreme plus a volatility buffer, bounded by a floor and a cap.	`SL = OB extreme ± 0.50 × ATR(14)` `Bounds: 18.0 pips ≤ SL ≤ 60.0 pips`
6	**Fixed target matrix** Enforces a constant 1 : 1.75 risk-to-reward.	`TP = Entry ± (SL distance × 1.75)`

Design rationale

Momentum, confirmed

A 20-period channel break marks directional expansion. Trend and slope filters remove entries against the prevailing move.

Exhaustion, filtered

The RSI window rejects breaks that have already run to extremes, where continuation odds deteriorate.

Invalidation, structural

The stop sits beyond the Order Block, the level whose failure invalidates the trade thesis, rather than at an arbitrary distance.

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SECTION 06

Risk Framework

Risk is set before the trade is placed and never adjusted after entry. The parameters below are fixed across all instruments and market regimes.

Parameter	Value	Architectural rationale
Risk Allocation	1.0% equity	Supports drawdown survival through 10+ consecutive losing trades.
Target Matrix	1 : 1.75 R	Breakeven win rate is 36.4%; profitable above that level.
Breathing Floor	18.0 pips	Protects against intra-bar noise and spread expansion at session rollover.
Volatility Cap	60.0 pips	Prevents oversized dollar risk during news-driven spikes.
Position Lock	1 trade / pair	Eliminates overlapping entries on correlated currency swings.

Payoff arithmetic

Expectancy per trade is `1.75 × p − (1 − p)`, where p is the win rate. Breakeven is `p = 1 / 2.75 = 36.4%`. Drawdown compounds as $1 - 0.99^{n}$. Both tables are arithmetic consequences of the parameters, not performance results.

Win rate	Expectancy per trade (R)	Reading
35%	-0.037 R	Below breakeven
40%	+0.100 R	Marginally positive
45%	+0.237 R	Positive
50%	+0.375 R	Positive
55%	+0.513 R	Positive

Consecutive losses	Equity drawdown at 1.0% per trade
5	4.90%
10	9.56%
15	13.99%
20	18.21%

Position sizing

Lot size is computed so that a stop-out costs exactly 1.0% of equity. As an **illustration only**: on $10,000 of equity, the risk budget is $100. A 25-pip stop on EURUSD, at roughly $10 per pip per standard lot, gives `$100 ÷ (25 × $10) = 0.40 lots`. The 18 to 60 pip stop range keeps the position size within a predictable band.

AGGREGATE EXPOSURE

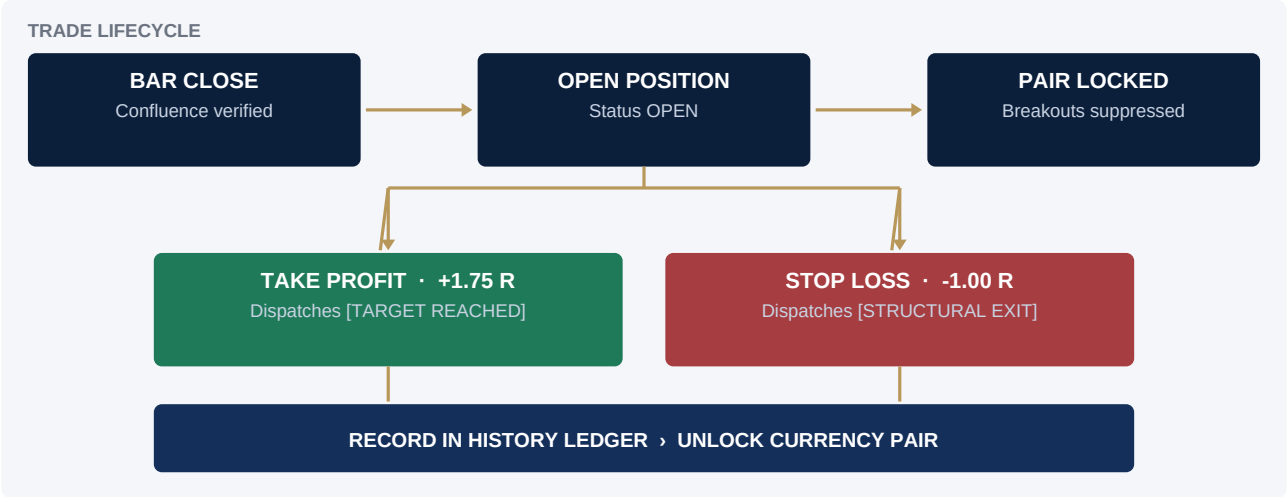
The one-trade-per-pair lock allows up to six simultaneous positions, or up to **6.0% of equity** at risk in total. Several pairs share USD exposure, so simultaneous positions are not fully independent. This is a stated property of the design.

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SECTION 07

Trade Lifecycle & Single-Trade Lock

To prevent duplicate exposure, over-trading and counter-trend churn, the engine enforces a **strict single-position lock per currency pair**. While a position is active, all further breakout triggers on that pair are suppressed across both 15M and 1H timeframes until the trade concludes.



Phase	Engine behavior	Ledger & interface state
Signal Trigger	A confirmed bar close satisfies all entry conditions. Exact lot size is computed for 1.0% equity risk.	Push alert sent to Telegram. Trade recorded as `OPEN`. Pair locked.
Surveillance	Breakout evaluation is bypassed for the locked pair. High/Low is monitored against SL and TP.	Shown in `/radar` with floating pips, entry, SL and TP.
Take Profit Hit	Candle extreme touches or exceeds TP. Realizes **+1.75 R**.	`[TARGET REACHED]` alert. Moved to history. Pair unlocked.
Stop Loss Hit	Candle extreme invalidates the Order Block boundary. Realizes **−1.00 R**.	`[STRUCTURAL EXIT]` alert. Moved to history. Pair unlocked.

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SECTION 08

Operator Interface

The Telegram terminal is engineered for mobile efficiency, using clean unbracketed numbering and compact 18-character dividers that prevent line wrapping on narrow screens.

Command	Scope	Function
`/radar`	Global · real-time	Surveillance radar separating active trades (floating pips, SL, TP) from monitored pairs (pips to breakout).
`/prices`	Spot quotes	Single-line quotes for EURUSD, USDJPY, USDCAD, USDCHF, AUDUSD and GOLD.
`/history`	`[pair] [range]`	Chronological trade journal with dynamic filters, e.g. `/history EURUSD 7d`.
`/performance`	`[pair]`	Win rate, realized pips, realized R and profit factor, strictly from trades on or after 2026-09-28.
`/strategy`	System rules	Formal algorithm specification, formulas and risk parameters.
`/status`	Infrastructure	Edge health, locked-position tally and cron operational state.

SECTION 09

Forward Verification Standard

Performance is reported under a strict **forward-testing standard**. Every metric shown by `/performance` is computed exclusively from live executions recorded in `trade_records.json` from **September 28, 2026** onward. No backtested results are published.

Metric	Definition
Win Rate (%)	`Winning Trades ÷ Total Closed Trades × 100`
Net Realized Pips	`Σ (Realized Exit Pips − Entry Pips), sign-adjusted for direction`
Net Realized Return (R)	$(+1.75 \times N_{wins}) - (1.00 \times N_{losses})$
Profit Factor	`Gross Winning Pips ÷ Gross Losing Pips`

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Interpreting a small sample

A win rate measured on few trades carries wide uncertainty. The table shows the one-standard-error margin around an observed 40% win rate, computed as $\sqrt{(p(1-p)/n)}$. Early forward results should be read with this in mind.

Closed trades	Approx. 1-sigma margin at 40%
30	± 8.9 pp
100	± 4.9 pp
300	± 2.8 pp

FORWARD AUDIT INTEGRITY
Forward tracking removes survivorship bias and historical lookahead. Each published metric is backed by a timestamped git commit of `trade_records.json`.

SECTION 10

Operational Resilience

Automated fail-safes are designed to prevent silently dropped alerts and to keep operation uninterrupted.

Fail-safe	Behavior
HTML Entity Auto-Fallback	If Telegram rejects formatting due to unescaped characters, the edge gateway strips markup and resends the alert as plain text.
Automatic Network Backoff	Timeouts and exponential retry on Yahoo Finance requests so transient network spikes do not abort scan cycles.
State Cache Pruning	Signal history in `market_scanner_state.json` is pruned to the last 200 IDs, bounding file growth while preserving duplicate-signal suppression.
Secret Handling	Token fallbacks accept both environment variables and repository secrets, preventing CI crashes when secrets are unset.

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SECTION 11

Limitations & Risk Disclosure

The design has known limits. They are stated here so that results can be interpreted in context.

Area	Limitation	Handling
Data quality	Yahoo Finance is a retail-grade feed, not an exchange-grade one. Candles can differ from a broker's prices.	Signals use completed bars only; retry and backoff on feed errors.
Scheduler timing	GitHub Actions scheduled runs are best-effort and can start late under platform load.	Confirmed-bar logic and duplicate-signal suppression tolerate delayed runs.
Platform dependency	The stack relies on free tiers of third-party services whose terms can change.	State is held in the repository, decoupled from any single runtime.
Intra-candle outcomes	Where one candle spans both SL and TP, OHLC data alone cannot show which was touched first.	Disclosed as a measurement limitation.
Execution costs	Spread, slippage and commissions reduce realized results relative to ledger R-multiples.	The 18.0 pip floor buffers against spread expansion.
Correlated exposure	Up to six positions may share USD risk, up to 6.0% of equity in aggregate.	One trade per pair; fixed 1.0% risk each.
Regime risk	Momentum breakouts tend to underperform in range-bound markets with repeated false breaks.	Trend, slope and RSI filters reduce, but do not remove, this exposure.
Sample size	The forward record begins on 2026-09-28. Early statistics carry wide uncertainty.	No performance is claimed before a meaningful sample exists.

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SECTION 12

Summary

The engine is a fully specified, serverless trading system whose rules, risk limits and verification method are all stated in advance.

Dimension	Position
Thesis	Disciplined, rule-based trading can run continuously without local hardware, recurring fees or opaque models.
Strategy	Donchian momentum breakouts confirmed by EMA trend and RSI filters, anchored to Order Block invalidation.
Risk	1.0% equity per trade, stops between 18.0 and 60.0 pips, fixed 1 : 1.75 target, one trade per pair.
Infrastructure	GitHub Actions, Cloudflare Workers and Telegram, at $0.00 per month.
Verification	Forward-only metrics from 2026-09-28, held in a timestamped git ledger.
Boundaries	Zero martingale, grid or averaging down. Known limits are disclosed in Section 11.

CURRENT STATUS
Forward audit began on **September 28, 2026**. This document reports no historical performance. Results will accumulate in the ledger and be reported through `/performance` as trades close.

SYSTEM ATTESTATION

Lead Quantitative Engineer

Status Active & forward-testing
Deployment GitHub Actions & Cloudflare Global Edge
Standard Proprietary & open forward audit
Published September 28, 2026

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APPENDIX A

Glossary

Term	Definition
Donchian Channel	Highest high and lowest low over a lookback window (here 20 periods); a close outside it marks a breakout.
EMA	Exponential moving average; weights recent prices more heavily. The engine uses the 20 and 50 period EMAs.
RSI	Relative Strength Index over 14 periods; a bounded momentum oscillator from 0 to 100.
ATR	Average True Range over 14 periods; a measure of typical price movement, used as a stop buffer.
Order Block	The last opposing candle before a displacement move, treated as an institutional accumulation or distribution zone.
R-multiple	Result expressed in units of initial risk. A stop-out is −1.00 R; a target hit is +1.75 R.
Profit Factor	Gross winning pips divided by gross losing pips.
Pip	Standard unit of forex price movement; 0.0001 for most pairs and 0.01 for JPY pairs.
Expectancy	Average R gained or lost per trade over many trades.

Important notice

This document describes a software system and does not constitute investment advice or an offer of any financial product. Trading leveraged foreign exchange and metals carries a high level of risk and may result in loss exceeding expectations. Parameter values and outcomes shown as arithmetic illustrations are not predictions or results. Past or hypothetical performance does not guarantee future results.

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